



CSC402

NEXA

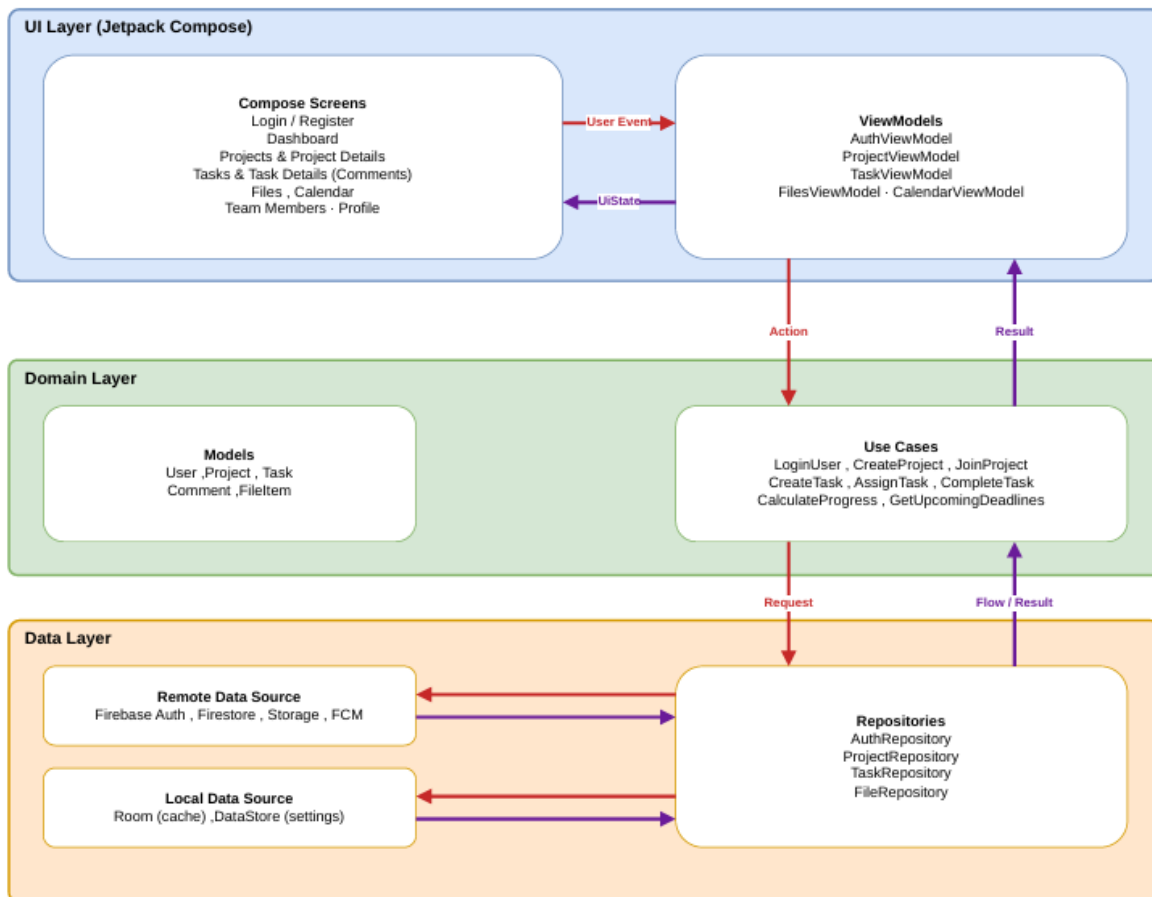
UI/UX Design & System Architecture

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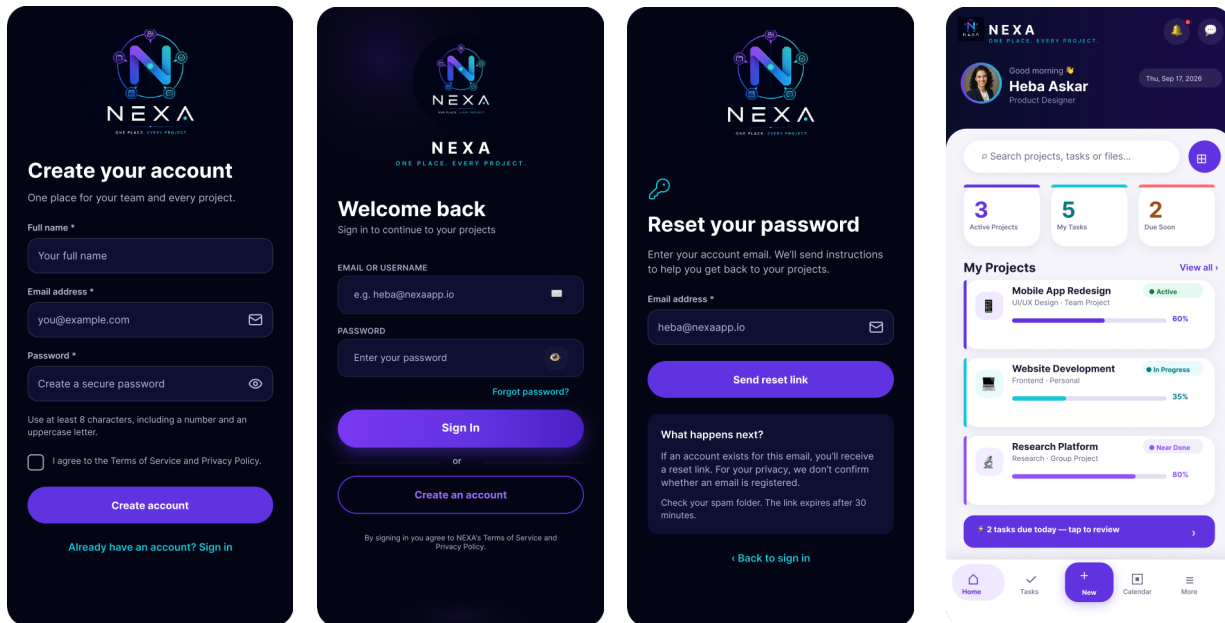
1. Application Architecture and Unidirectional Data Flow

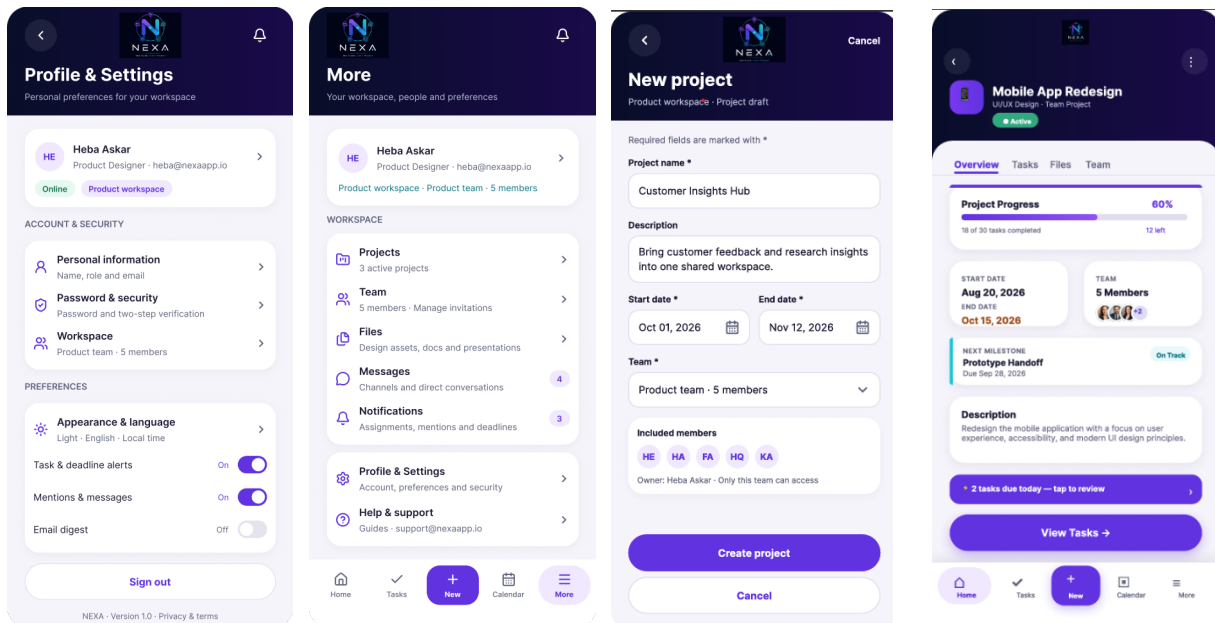
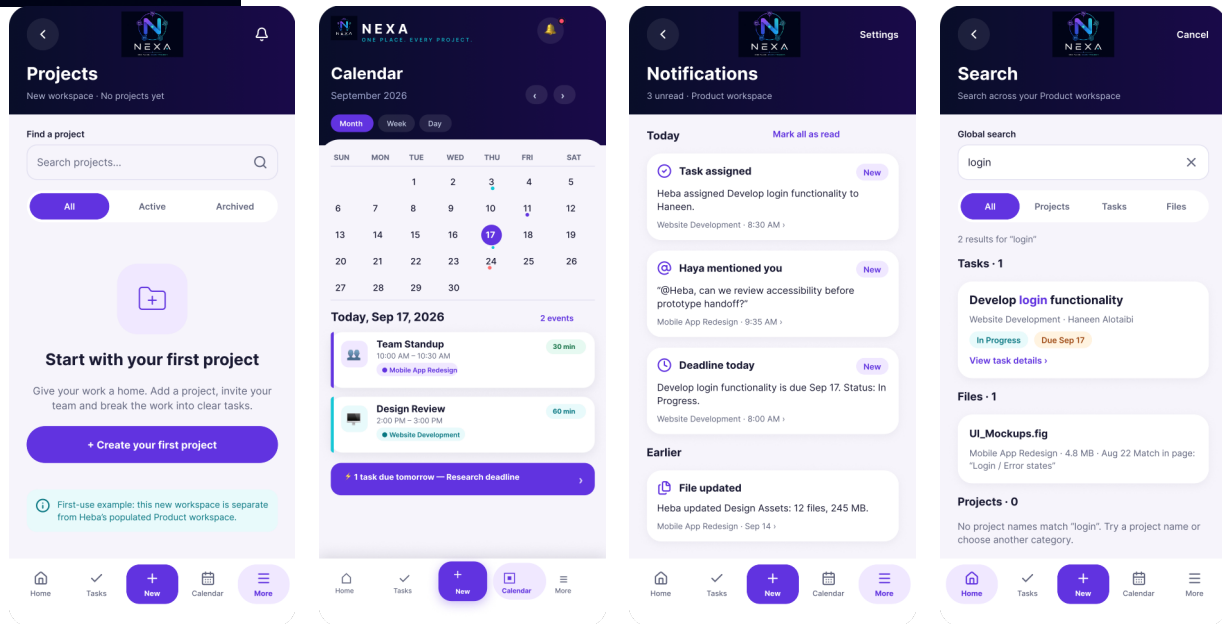
NEXA – App Architecture

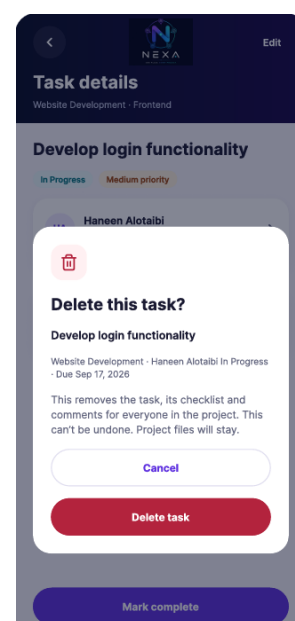
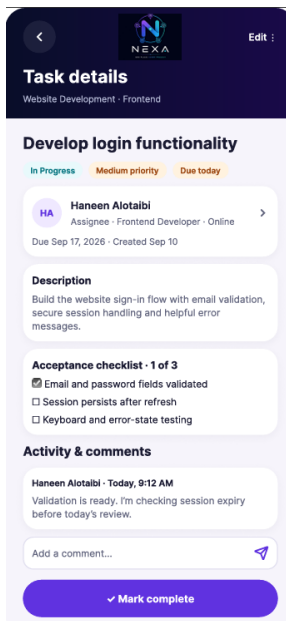
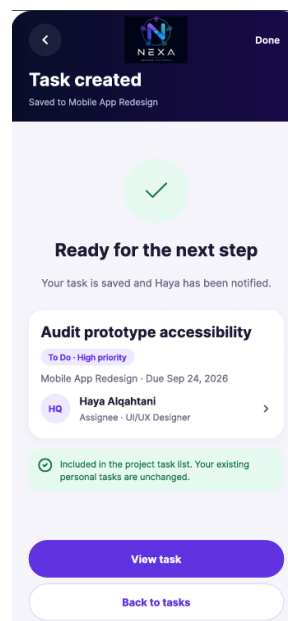
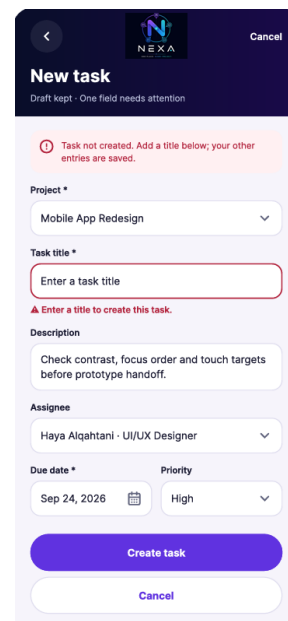
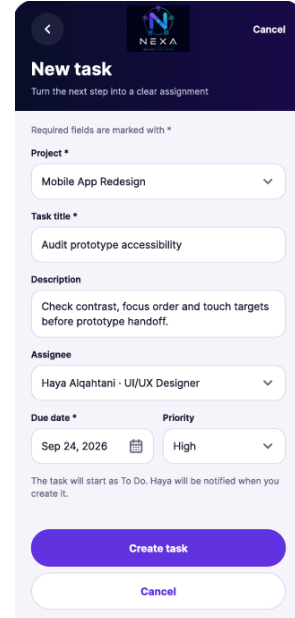
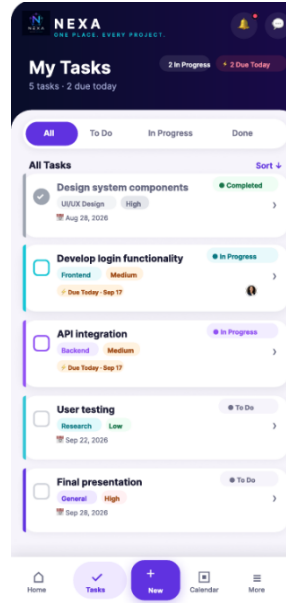
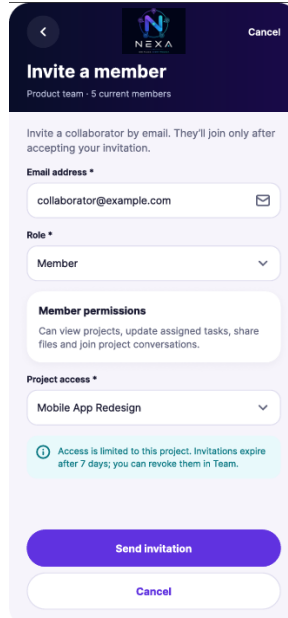
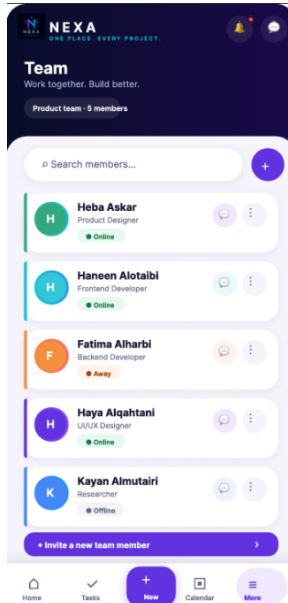


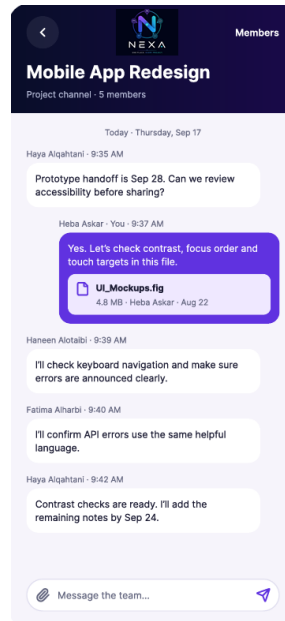


2. UI/UX Design & App Wireframes



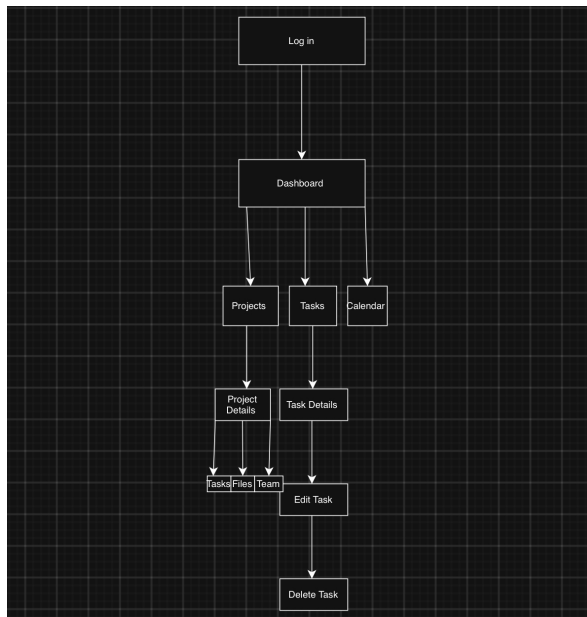








3. App Navigation Map



4. Jetpack Components and Libraries with Justification

NEXA will use the following Jetpack components and supporting libraries to implement its features. The selection aligns with the application's UI, Domain, and Data layers and supports Unidirectional Data Flow (UDF).

Component / Library	Planned Use in NEXA	Justification
Jetpack Compose	Build screens such as Login, Dashboard, Projects, Tasks, and Team Members.	Enables reusable UI components and updates the interface when the observed application state changes.



ViewModel	Manage screen state and handle user actions through ViewModels such as AuthViewModel, ProjectViewModel, and TaskViewModel.	Separates screen logic from composables and retains UI state across configuration changes.
Room Database	Cache project and task data locally.	Supports structured local queries and access to previously cached data without an internet connection. Repositories will coordinate updates between Room and Firestore.
DataStore	Store local user settings and preferences.	Provides asynchronous persistence for small settings, keeping user preferences available between app launches.
Firebase Authentication	Support account registration, login, logout, and password reset.	Provides managed authentication without requiring the team to develop a custom password authentication service.
Cloud Firestore	Store shared projects, tasks, memberships, deadlines, and task comments.	Supports real-time data updates so team members can observe changes to tasks and project progress.
Cloud Storage for Firebase	Store and retrieve project documents and attachments.	Handles file contents separately from structured project data and supports access control through security rules.
Firebase Cloud Messaging (FCM)	Receive notifications about task assignments, updates, and deadlines.	Enables push notification delivery from a trusted backend. Deadline scheduling and notification triggers require separate backend logic.